

Developing and Using Models Both scientists and engineers use models, such as those shown in **Figure 1-10**. When you hear the word *model*, you may think of model trains or other representations of much larger objects. But there are many different kinds of models. Mathematical representations and computer simulations are examples, as are two-dimensional drawings, diagrams, and maps. In fact, the flowcharts used throughout this lesson are models. Models help visualize and summarize ideas and communicate ideas to others.

Using Mathematics and Computational Thinking

Mathematics and computational thinking (the process of mathematical calculation) are also important tools. The relationships between variables are essential to the understanding of scientific phenomena. Ratios, rates, percentages, and unit conversions are used to analyze and interpret data. A mathematical representation can represent and model data; it can also support claims and explanations.

Constructing Explanations and Designing Solutions

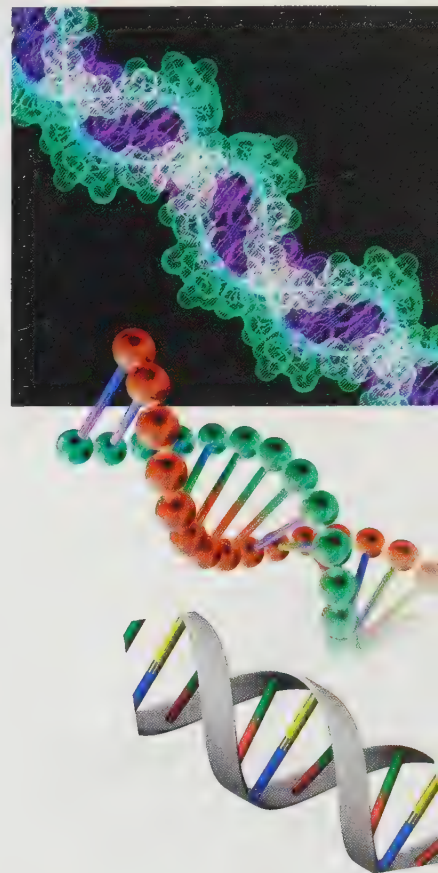
Scientists aim to construct explanations for events in the natural world that caused them to ask a question. After a scientist collects data, analyzes and interprets that data, and finds that it supports the hypothesis, the next step is to construct an explanation. An explanation describes how variables relate to one another, and it is supported by evidence. In engineering, the goal is to design a solution to a problem within certain constraints.

Engaging in Argument From Evidence In science, an argument is a reason or set of reasons used to persuade others that an idea is right or wrong. Once a scientist has constructed an explanation, the next step is to use evidence (analyzed and interpreted data) to persuade others that the explanation is correct. Part of this process is receiving critiques and responding thoughtfully. Engineers defend claims that a design solution is effective or evaluate competing design solutions.

Figure 1-10

Models

Models can be elaborate pictures that a computer generates (top), physical models, (middle), or illustrations (bottom). All three of these models show DNA, a molecule essential for life.



LESSON 1.2 Review

KEY QUESTIONS

1. How are both curiosity and skepticism useful in science?
2. How can peer review help scientists improve their work?
3. How is the use of science related to its context in society?
4. Why is testing ideas an important part of all science and engineering practices?

CRITICAL THINKING

5. **Apply Concepts** An advertisement claims that studies of a new sports drink show it boosts energy. You discover that none of the study results have been peer-reviewed. What would you tell consumers who are considering buying this product?
6. **Apply Concepts** A study shows that a new pesticide is safe for use on food crops. The researcher who conducted the study works for the pesticide company. What potential biases may have affected the study?